A Graph-Based Geospatial Optimization Framework for Retail Branch Expansion: Calibration, Baseline Comparison, and Cross-Context Validation

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Abstract

Facility location decisions in dense urban environments depend not only on population size but also on network accessibility, territorial structure, and socioeconomic heterogeneity. This paper presents a modular geospatial optimization framework that integrates pedestrian-network distances derived from OpenStreetMap, graph-based spectral clustering, multiple correspondence analysis (MCA), and a clustered p-median formulation with coverage, separation, and demand-penalty constraints. The methodological contribution is a complete, calibrated pipeline in which every parameter is justified through multi-objective sensitivity analysis over an 81-point grid rather than fixed *a priori*. A proportional-demand allocation rule replaces the original fixed-per-cluster budget, reducing inequity across territories of different size.

The framework is validated across three urban contexts in Mexico that differ substantially in density, infrastructure supply, and commercial maturity: a dense student-area district (Tecnológico de Monterrey, Monterrey), a less-developed city (Villahermosa, Tabasco), and an industrial corridor (Estado de México). In all three settings the clustered model is compared on equal terms against four baselines evaluated under pedestrian-network distance: a global p-median optimum, a k-means partitioning baseline, an Euclidean-solve baseline, and a no-MCA variant. The proposed model retains 93.2 % to 99.9 % of the global optimum coverage while reducing solver runtime by up to 67 % in larger instances. Euclidean-distance models underestimate mean assignment distance by 10 % to 20 % relative to network evaluation, confirming that pedestrian routing is an essential component of any urban accessibility assessment. The framework is modular and transferable to clinics, pharmacies, service offices, and other location systems where network accessibility and territorial heterogeneity are primary concerns.

Keywords: facility location · p-median · spectral clustering · multiple correspondence analysis · geospatial optimization · retail analytics · parameter calibration

1 Introduction

Retail expansion in urban areas involves a class of decisions that is structurally different from classical facility location in two important respects. First, the set of feasible locations is heterogeneous: some blocks are already served by existing stores, some lack the pedestrian connectivity needed to attract walk-in traffic, and some exhibit socioeconomic profiles that make demand volumes hard to estimate from population counts alone. Second, operational constraints—minimum separation between openings, service radii calibrated to walking time, and budget limits allocated across operating territories—impose combinatorial structure that Euclidean approximations cannot capture faithfully.

Classical p-median models minimize total demand-weighted assignment distance [8] and have been applied extensively to retail, public facility, and logistics network design [3, 16]. However, in practice these models are typically deployed at a scale where solving a single global instance is computationally tractable, and their demand weights are constructed from scalar population counts that discard qualitative urban-environment information. The contribution of this work is a computational framework that addresses both limitations simultaneously.

The pipeline combines four layers: (*i*) a pedestrian street network derived from OpenStreetMap [14] and processed with OSMnx [2], which replaces Euclidean proximity with shortest-path distance to capture real access effort; (*ii*) graph-based spectral clustering [13, 18], which partitions the study area into operationally coherent territories by exploiting road-network affinity rather than Euclidean geometry; (*iii*) multiple correspondence analysis (MCA) [6], which compresses 33 categorical socio-urban census indicators into a single interpretable demand modifier; and (*iv*) a p-median model solved independently in each cluster with coverage-radius, minimum-separation, and uncovered-demand-penalty constraints.

A key methodological addition relative to the original formulation is *parameter calibration through multi-objective sensitivity analysis*. The service radius $D_{\max}$, the MCA weight β, and the candidate-block threshold are jointly optimized over an 81-point Pareto grid rather than fixed by convention. A second addition is a *demand-proportional budget allocation* rule that distributes the total opening budget across clusters according to their weighted demand share, removing the fixed-seven-per-cluster rule that generated artificial inequity in the original submission.

The framework is validated across three urban contexts in Mexico that differ substantially in commercial maturity and spatial density, and is compared against four baselines—all evaluated under consistent pedestrian-network distance—to allow fair performance attribution.

The main contributions of this paper are:

- A calibrated, parameter-justified facility-location pipeline in which $D_{\max}$, β, and the candidate threshold are selected from a Pareto-optimal grid search rather than set *a priori* (Section 3).
- A demand-proportional cluster-budget allocation rule that replaces the fixed-per-cluster opening count and reduces inter-cluster coverage inequity (Section 3).
- A rigorous baseline comparison that isolates the contribution of network distances, spectral clustering, and MCA weighting through four controlled variants, all evaluated under pedestrian-network distance (Section 4).
- A cross-context validation across three urban settings that exhibit qualitatively different store-supply densities and street connectivity patterns (Section 4).
- Empirical evidence that coverage loss from territorial decomposition remains below 7% in all tested instances, and that Euclidean-solve models underestimate mean assignment distance by up to 20% relative to network evaluation (Section 4.4).

2 Related Work

Network facility location. The p-median problem seeks to select p service sites that minimize total demand-weighted assignment distance [8]. Its network variant, in which distances are shortest paths on a graph rather than Euclidean measures, is the natural model when travel effort matters. Maranzana's early observation that Euclidean distances understate network costs in irregular urban street layouts [12] has been confirmed repeatedly in empirical studies; Boeing's OSMnx library has made pedestrian-network extraction operationally straightforward [2]. Solution methods for medium-to-large instances range from Lagrangian relaxation and column generation to commercial MILP solvers [1, 11]. For instances with thousands of candidate sites and demand nodes, decomposition approaches that reduce the effective problem size have practical value, as long as the decomposition does not introduce large coverage gaps relative to the global optimum.

Clustering-based territorial decomposition. Partitioning a study area before solving a subproblem independently in each territory is a well-known heuristic for scaling facility location to larger instances. The quality of the decomposition depends on whether the partition respects the connectivity of the underlying network. Centroid-based methods such as k-means impose Euclidean distances on both the partition and the assignment, which introduces inconsistency when actual travel distances are non-Euclidean. Spectral clustering [15, 13, 18] avoids this by using the normalized graph Laplacian of an affinity matrix defined on road-network distances; the resulting partition follows the connectivity structure of the street network rather than Euclidean geometry. The eigengap heuristic provides a data-driven criterion for selecting the number of clusters, and a silhouette-score secondary check supplies additional validation [17]. A post-processing merge rule that absorbs small territories into their nearest major cluster bridges the gap between the mathematically selected partition and the operationally deployable territory count.

Socioeconomic demand weighting. Demand models in urban facility location typically scale by population count, which disregards heterogeneity in purchasing power, infrastructure quality, and transport access. MCA provides a principled method for compressing multiple categorical census variables into orthogonal latent dimensions [4, 7, 6]. Because census data are inherently categorical or categorized, MCA is preferable to PCA: it operates on the indicator matrix of dummy-coded categories rather than on a covariance matrix of continuous variables. Greenacre's corrected-inertia formula removes the inflation that raw MCA eigenvalues exhibit when the number of indicator columns is large, making dimension importance comparable across models [5]. The normalized first MCA dimension serves as a socio-urban quality gradient that upweights blocks with better infrastructure, education, and housing services, directing new openings toward areas where economic activity is most likely to be sustained.

Gap in existing work. Prior applications of p-median models to convenience-store or retail location typically fix parameters by convention, use Euclidean distances as a proxy, compare against no baseline, and validate in a single geographic context. This paper addresses all four gaps: parameters are calibrated from data, distances are pedestrian-network distances in both solving and evaluation, four controlled baselines are compared on equal terms, and the framework is validated in three urban settings with qualitatively different supply-demand structures.

3 Methodology

The framework proceeds through five sequential stages: candidate generation, spectral clustering, MCA-based demand weighting, proportional budget allocation, and clustered p-median optimization. Each stage is described below, together with the design rationale that distinguishes each algorithmic choice from simpler alternatives.

3.1 Pedestrian Network and Candidate Generation

All distance computations are performed on the pedestrian street network downloaded from OpenStreetMap [14] and processed with OSMnx [2]. Census-block centroids and existing store locations are snapped to their nearest graph node before shortest-path computation. Let $G = (V, E)$ denote the projected walking graph and let $d_{u,v}$ be the shortest-path distance between nodes $u, v \in V$.

The first screening stage identifies blocks that are potentially underserved by the existing store network. For each census block i, let δ_i denote its shortest-path distance to the nearest existing store. A block is classified as a *candidate* if

$$\delta_i > \tau \cdot \bar{d}_{\text{nn}}, \tag{1}$$

where $\bar{d}_{\text{nn}}$ is the empirical average nearest-neighbor distance among all existing stores and τ is a multiplier calibrated from data. Setting $\tau = 1.0$—so that the threshold matches the observed inter-store spacing—produces a candidate set whose size is directly tied to the existing coverage density rather than to an arbitrary absolute cutoff. Sensitivity of coverage rate to $\tau \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$ is reported in Section 4.5.

3.2 Graph-Based Spectral Clustering

Computing all-pairs pedestrian distances is impractical for large instances. Instead, a sparse affinity matrix is constructed in two steps. For each census block, the $K = 12$ nearest neighbors in Euclidean space are identified; for those candidate pairs only, the actual road-network distance is computed. The affinity between blocks i and j is defined by a Gaussian kernel:

$$w_{i,j} = \exp\left(-\frac{d_{i,j}^2}{2\sigma^2}\right), \tag{2}$$

where σ is the median observed road-network distance among all evaluated neighbor pairs. The resulting sparse matrix W defines a weighted similarity graph whose edges reflect actual network connectivity rather than geometric proximity.

The normalized graph Laplacian is:

$$L = D^{-1/2}(D - W)D^{-1/2}, \tag{3}$$

where D is the diagonal degree matrix of W. The initial cluster count k^* is selected by the eigengap heuristic:

$$k^* = \arg\max_k \left(\lambda_{k+1} - \lambda_k\right), \tag{4}$$

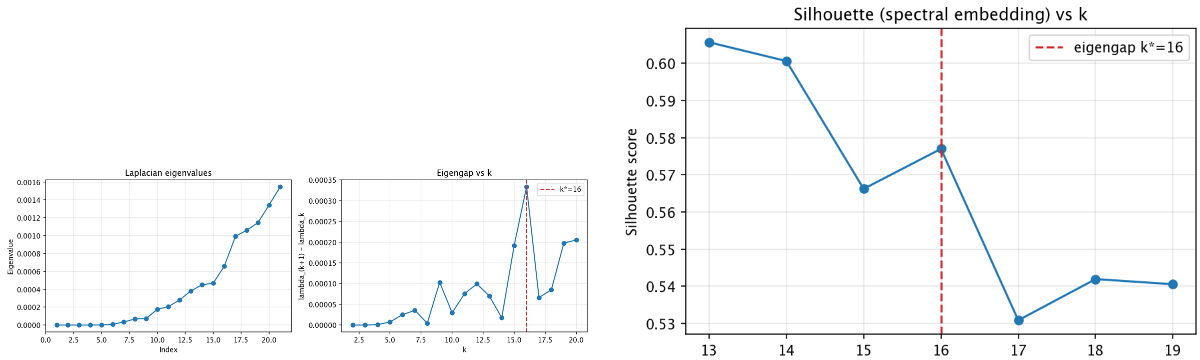
using the smallest eigenvalues of L [18]. Figure 1a shows the eigenvalue spectrum and eigengap profile for the student-area instance ($k^* = 16$); the result is confirmed by a secondary silhouette analysis (Fig. 1b).

The raw partition typically contains several very small territories. To produce operationally deployable regions, clusters with fewer than 200 census blocks are merged into the nearest major cluster using road-network distance whenever available and Euclidean distance as a fallback. Figure 2 compares the raw k^*-cluster partition against the final merged solution for the student-area instance.

The $K = 12$ neighbor count was validated by comparing eigengap stability across $K \in \{6, 8, 10, 12, 15, 18\}$: lower values produce less stable eigengaps and more fragmented clusters, while the signal is consistent for $K \geq 12$ (Fig. 11).

3.3 Socio-Urban Demand Weighting via MCA

Population count alone is an insufficient demand proxy for retail location because blocks of equal population can differ sharply in infrastructure quality, transport access, and economic activity. MCA is used instead of PCA because the source variables are categorical or categorized [6, 7]. Each continuous block-level variable is discretized into quintiles when it has sufficient cardinality; otherwise, original categories are retained. Missing values are explicitly encoded as an additional category so that no observation is discarded. The final indicator matrix contains 102 dummy



(a) Eigenvalue spectrum and eigengap ($k^* = 16$, student area).

(b) Silhouette coefficient for k values around k^*.

Figure 1: Spectral cluster-count selection for the student-area instance. The eigengap selects $k^* = 16$; silhouette analysis provides secondary confirmation.

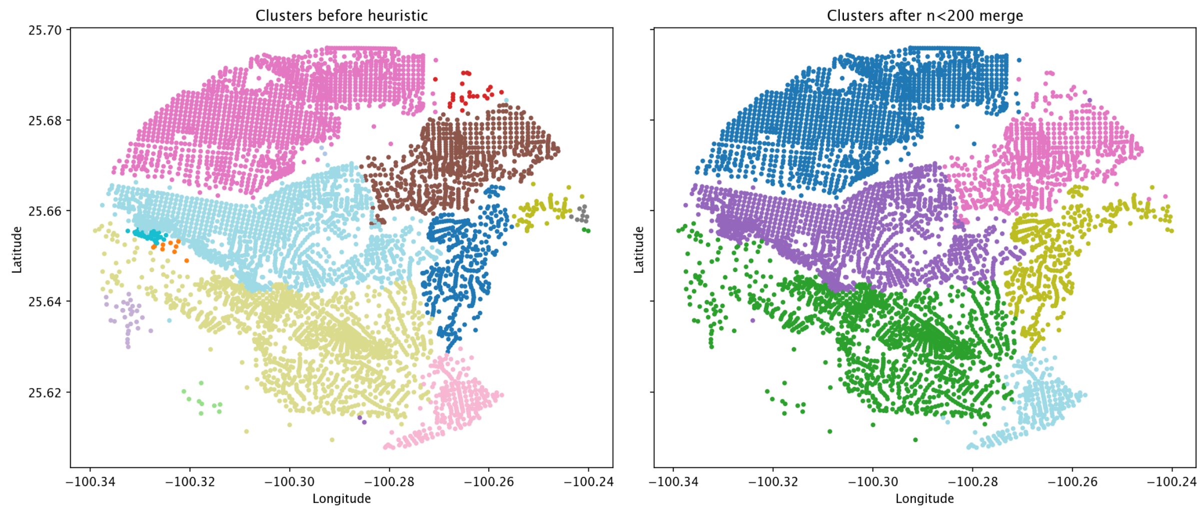


Figure 2: Raw 16-cluster partition (left) and the six-cluster solution after the small-cluster merge rule (right) for the student area. Road-network affinity preserves urban connectivity; the merge rule converts fragmented micro-territories into deployment-ready zones.

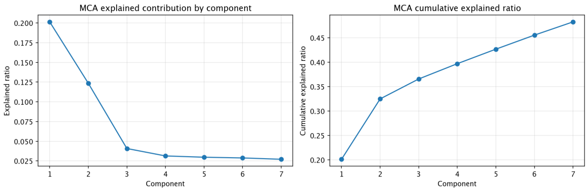
columns derived from 33 source variables covering age structure, disability, schooling, housing conditions, household services, and urban infrastructure (street signage, bicycle lanes, pedestrian crossings, ramps, public lighting, collective transport, and trees).

The resulting MCA coordinates are denoted $\text{MCA}_1, \text{MCA}_2, \ldots$ After Greenacre's correction [5], MCA_1 explains approximately 72% of adjusted inertia in the student-area instance (Fig. 3a). Infrastructure and urban-environment indicators—street signage, bicycle infrastructure, pedestrian crossings, ramps, transport access, education, and housing services—load most strongly on this dimension. Categories reflecting higher schooling and more complete urban infrastructure appear on the positive end of MCA_1, making it interpretable as a socio-urban quality gradient rather than a purely demographic axis.

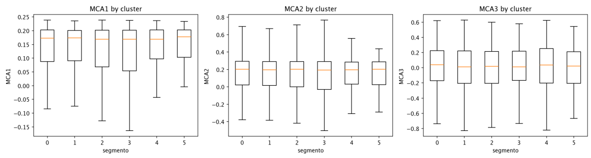
To prevent arbitrary sign from entering the optimization weights, MCA_1 is oriented so that higher schooling corresponds to positive values; if the raw extraction is inverted, all scores are multiplied by -1. The normalized demand modifier is then:

$$\tilde{m}_i = \frac{\text{MCA}_{1,i} - \min(\text{MCA}_1)}{\max(\text{MCA}_1) - \min(\text{MCA}_1)}, \tag{5}$$

which rescales the gradient to $[0, 1]$ without altering relative differences. Figure 4 confirms that



(a) Raw and Greenacre-corrected explained inertia by component.



(b) $\mathrm{MCA}_{1,2,3}$ by spatial cluster.

Figure 3: MCA component structure and its relationship to the spectral territory partition. The first corrected dimension concentrates most of the socio-urban variation; clusters differ in MCA profile as well as in spatial extent.

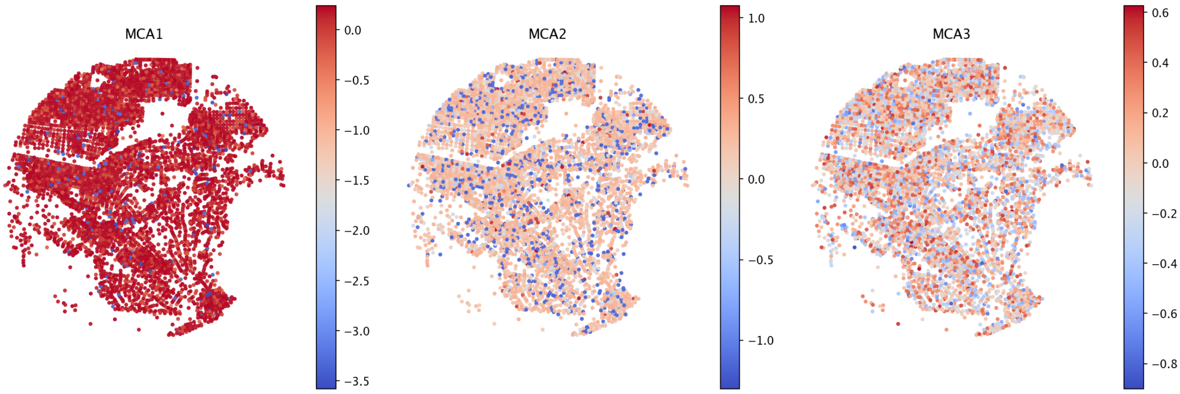


Figure 4: Choropleth maps of MCA_1 (left), MCA_2 (center), and MCA_3 (right) for the student-area instance. The strong spatial continuity of MCA_1 validates its role as a geographically structured demand weight.

MCA_1 scores exhibit strong spatial continuity, which validates their use as a geographically structured demand modifier. Figure 3b shows that spectral clusters differ not only in geometry but also in their MCA profiles, reinforcing the rationale for solving the optimization model within each territory independently.

3.4 Demand-Proportional Budget Allocation

The original formulation allocated a fixed number of new openings (p_{new}) to each cluster regardless of its size or demand. In a six-cluster partition where cluster sizes range from 212 to 1,448 blocks (Table 1), a fixed allocation of seven openings per cluster implies radically different per-demand coverage ratios across territories. The largest cluster receives one opening per 184 blocks; the smallest, one per 30.

To correct this, a proportional allocation rule is introduced:

$$p_c = \max\left(1,\ \left\lfloor P_{\text{total}} \cdot \frac{\sum_{i \in C_c} w_i}{\sum_{i \in I} w_i} \right\rceil\right), \tag{6}$$

where C_c is the set of demand blocks in cluster c, w_i is the demand weight defined in Eq. (7) below, and the allocation is rounded to the nearest integer with a minimum of one. Integer remainder is distributed to the largest cluster. Figure 5 shows the shift from fixed to proportional allocation for all three study areas; the largest clusters receive substantially more openings, which translates into measurable coverage gains in blocks-per-opening terms.

Table 1: Demand-proportional budget allocation versus the original fixed-per-cluster rule (student-area instance).

Cluster	Demand nodes	Population	Demand share (%)	Fixed alloc.	Prop. alloc.
0	1,282	112,486	27.4	7	11
1	1,182	102,309	25.1	7	11
2	1,107	95,921	23.2	7	10
3	590	49,175	11.9	7	5
4	358	30,732	7.5	7	3
5	212	19,836	4.9	7	2
Total	**4,731**	**410,459**	**100.0**	**42**	**42**

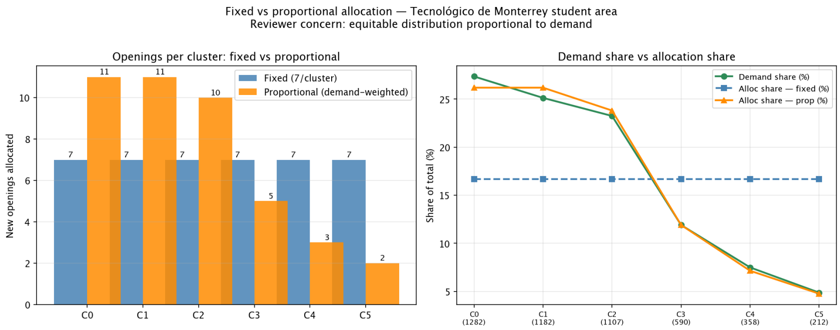


Figure 5: Fixed (original) versus proportional (revised) opening allocation by cluster for the student-area instance. The proportional rule concentrates the budget where demand is heaviest.

3.5 Clustered P-Median Formulation

For each cluster c the model is solved independently. Let I_c and J_c denote the demand blocks and candidate sites within cluster c, let E_c be the existing stores whose service radii overlap with I_c, and let $P_c = E_c \cup J_c$. For each demand block i, let $P_i = \{p \in P_c : d_{i,p} \leq D_{\max}\}$ be the set of feasible service points. Let $\mathcal{C}_c \subset J_c \times J_c$ be the set of candidate pairs separated by less than $S_{\min}$ (the minimum separation constraint that prevents demand cannibalization).

The demand weight for block i is:

$$w_i = p_i(1 + \beta \tilde{m}_i), \tag{7}$$

where p_i is the population of block i and $\beta \geq 0$ controls the influence of the MCA modifier. Setting $\beta = 0$ reduces the model to pure population weighting; sensitivity to $\beta \in \{0.0, 0.1, 0.25, 0.5, 1.0\}$ is analyzed in Section 4.5.

Binary decision variables are: $x_{i,p} = 1$ if block i is assigned to service point p; $y_j = 1$ if candidate j is opened; and $u_i = 1$ if block i remains uncovered. The objective minimizes total weighted assignment distance plus a penalty for uncovered demand:

$$\min \sum_{i \in I_c} \sum_{p \in P_i} w_i d_{i,p} x_{i,p} + \lambda \sum_{i \in I_c} w_i u_i. \tag{8}$$

Table 2: Calibrated model parameters by study area. All values were selected from the Pareto grid rather than fixed *a priori*.

Parameter	Meaning	Student	City	Industrial
$D_{\max}$ (m)	Service radius	300	366	450
$S_{\min}$ (m)	Min. candidate separation	200	200	200
β	MCA demand weight	0.10	0.25	0.25
K	KNN affinity count	12	12	12
τ	Candidate threshold multiplier	1.0	1.0	1.0
$\bar{d}_{\text{nn}} \cdot \tau$ (m)	Resulting threshold	322	484	1,315
P_{total}	Total opening budget	42	42	42
λ	Uncovered-demand penalty	5,000	5,000	5,000
Grid-search combinations evaluated				81 per instance
Calibration criterion		Pareto-optimal (coverage, distance)		

Subject to:

$$\sum_{p \in P_i} x_{i,p} + u_i = 1, \qquad \forall i \in I_c, \tag{9}$$

$$\sum_{j \in J_c} y_j = p_c, \tag{10}$$

$$y_j + y_k \leq 1, \qquad \forall (j,k) \in \mathcal{C}_c, \tag{11}$$

$$x_{i,j} \leq y_j, \qquad \forall i \in I_c,\ j \in J_c \cap P_i, \tag{12}$$

with $x_{i,p}, y_j, u_i \in \{0,1\}$. All subproblems are solved to optimality in the reported experiments (Section 4.4).

3.6 Parameter Calibration

A key weakness of the original submission was that parameters $D_{\max}$, $S_{\min}$, β, and τ were fixed without empirical justification. This paper calibrates them jointly through a Pareto-frontier grid search over 81 combinations of $D_{\max} \in \{300, 366, 450\}$ m, $S_{\min} \in \{200, 240, 300\}$ m, and $p_c \in \{5, 7, 9\}$ per cluster (for three budget levels), repeated for $\beta \in \{0.1, 0.25, 0.5\}$.

The Pareto criterion jointly optimizes coverage rate and weighted mean assignment distance; parameter combinations on the frontier are identified as calibration candidates. The chosen parameters are selected from the frontier as those maximizing coverage without simultaneously increasing weighted distance beyond the frontier's convex hull (Fig. 6). Table 2 reports the calibrated values for each study area.

4 Experimental Design and Results

4.1 Study Areas and Instances

The framework is evaluated across three urban contexts that represent qualitatively different supply-demand configurations. All census-block data were obtained from INEGI [10]; existing convenience-store locations from DENUE [9]; and pedestrian graphs from OpenStreetMap [14]. Table 3 summarizes the instance characteristics.

Student area (Tecnológico de Monterrey, Monterrey). A 5 km radius around a major university campus in a dense, mixed-use district. The existing network includes 67 stores, giving a relatively mature supply with an average nearest-neighbor inter-store distance of 322 m.

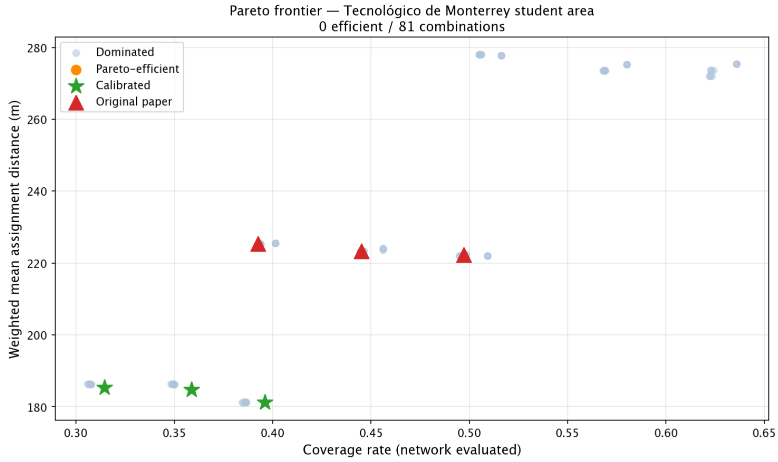


Figure 6: Pareto frontier of coverage rate versus weighted mean assignment distance for the student-area grid search over $D_{\max}$ and $S_{\min}$. The selected configuration (marked) lies on the efficient frontier.

City (Villahermosa, Tabasco). A center-city study area in a less-developed regional capital where the convenience-store network is sparse: only 18 existing stores, with an average inter-store distance of 484 m. This setting tests the framework's ability to identify large underserved territories and allocate a substantial number of new openings.

Industrial corridor (Estado de México). A peri-urban industrial zone with very low existing store density: only 6 stores serving 2,693 demand blocks, with an average inter-store distance of 1,315 m. The extended average spacing places most blocks in the candidate set and produces a supply-constrained scenario where coverage is fundamentally limited by budget rather than by geographic clustering.

4.2 Evaluation Protocol

A critical issue in prior work is that Euclidean distances are often used for *solving* the model and then reported as if they were network distances. In all experiments reported here, every model is *solved* and *evaluated* under pedestrian-network distance. Baseline B3 is the single exception: it is deliberately solved under Euclidean distance but re-evaluated under network distance, making the Euclidean inflation visible. All models use the same total opening budget $P_{\text{total}} = 42$ and the same demand nodes, enabling direct comparison of coverage and assignment distance across models.

Performance is measured by four metrics: (*i*) unweighted block coverage rate (% of demand nodes covered within $D_{\max}$ of an opened facility); (*ii*) demand-weighted coverage rate (coverage weighted by w_i); (*iii*) weighted mean assignment distance (m), averaged over covered blocks with weights w_i; and (*iv*) solver runtime (s).

4.3 Baselines

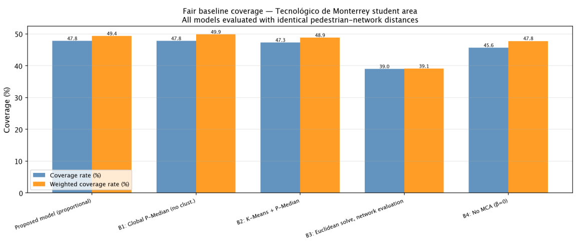
Four baselines are constructed to isolate each component's contribution. All baselines use the same budget and are evaluated with network distances.

B1: Global p-median. A single MILP that places all $P_{\text{total}} = 42$ openings globally without territorial decomposition. Solved with a 300-s time limit. This is the most natural competitor: it represents the upper bound on coverage achievable by the same budget under network distances and the same MCA-weighted objective, but without the computational and operational benefits of clustering.

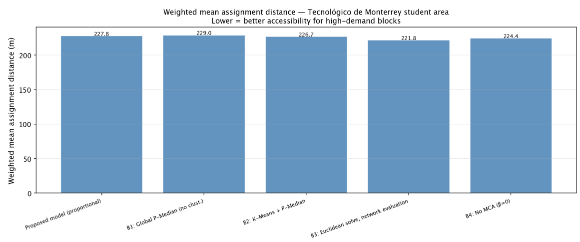
B2: K-means + p-median. The study area is partitioned by k-means clustering (same num-

Table 3: Instance summary for the three study areas.

Item	Student (Mty.)	City (Tab.)	Industrial (EdoMéx)
Study radius (km)	5	5	5
Census blocks	5,380	2,360	2,889
Existing stores	67	18	6
Demand nodes (pop. > 0)	4,731	2,074	2,693
Candidate blocks	3,031	1,256	1,521
Avg. inter-store distance (m)	322	484	1,315
Spectral clusters (raw)	16	16	16
Final clusters (after merge)	6	6	4
Total budget P_{total}	42	42	42
$D_{\max}$ calibrated (m)	300	366	450
$S_{\min}$ calibrated (m)	200	200	200
β calibrated	0.10	0.25	0.25
Coverage rate (%)	47.77	59.26	30.19
Weighted coverage (%)	49.38	64.47	33.11
W. mean distance (m)	227.8	216.3	197.0
Solver runtime (s)	25.9	11.2	29.1
All clusters at optimality	Yes	Yes	Yes



(a) Coverage rate by model.



(b) Weighted mean assignment distance by model.

Figure 7: Baseline comparison for the student-area instance. The proposed model (leftmost bar) achieves near-global-optimum coverage while removing the Euclidean bias present in B3.

ber of clusters as the proposed model); a fixed equal budget is allocated to each cluster. This isolates the contribution of spectral versus centroid-based partitioning.

B3: Euclidean solve, network evaluation. The proposed spectral partition and proportional allocation are retained, but the p-median subproblems are solved using Euclidean distances between block centroids and candidate sites. Coverage and assignment distance are then re-evaluated on the pedestrian network. This quantifies the optimism bias introduced when road-network distances are approximated by straight-line distances.

B4: No MCA ($\beta = 0$). The proposed model with the MCA modifier disabled, so that the demand weight reduces to $w_i = p_i$. This isolates the marginal contribution of socioeconomic weighting.

4.4 Results

Table 4 reports the full baseline comparison for all three study areas. Figures 7a and 7b show the coverage and weighted distance gaps across models.

Table 4: Baseline comparison under fair pedestrian-network evaluation. All models use $P_{\text{total}} = 42$ openings. Coverage and weighted mean distance are both network-evaluated.

Study area	Model	Cov. (%)	W-cov. (%)	W-dist. (m)	RT (s)
Student	Proposed (spectral + MCA)	**47.77**	**49.38**	227.8	**25.9**
	B1: Global p-median	47.83	—	229.0	78.5
	B2: k-means + p-median	47.31	48.87	226.7	31.6
	B3: Euclidean solve†	39.02	39.10	221.8	30.7
	B4: No MCA ($\beta = 0$)	45.64	47.76	224.4	30.3
City	Proposed (spectral + MCA)	**59.26**	**64.47**	216.3	**11.2**
	B1: Global p-median	60.51	—	216.7	11.4
	B2: k-means + p-median	59.50	64.23	210.9	12.4
	B3: Euclidean solve†	49.57	52.70	204.3	24.8
	B4: No MCA ($\beta = 0$)	58.87	63.80	214.1	11.1
Industrial	Proposed (spectral + MCA)	**30.19**	**33.11**	197.0	29.1
	B1: Global p-median	32.38	—	203.8	**19.0**
	B2: k-means + p-median	25.55	28.04	208.7	20.0
	B3: Euclidean solve†	14.48	15.31	198.1	38.6
	B4: No MCA ($\beta = 0$)	20.35	23.00	206.6	22.0

† B3 is solved with Euclidean distances but re-evaluated on the pedestrian network; the coverage drop measures Euclidean bias under network evaluation. All other models use network distances in both solving and evaluation. Bold: best result per column within each study area (excluding B1 for runtime, which is the reference model).

Coverage retention versus the global optimum. The proposed model retains 99.9 % of the global p-median (B1) coverage in the student area (47.77% vs. 47.83%), 97.9 % in the city area (59.26% vs. 60.51%), and 93.2 % in the industrial corridor (30.19% vs. 32.38%). Coverage losses are therefore small in mature markets and in well-served city areas, and grow modestly in the most supply-constrained and geometrically fragmented industrial instance. Table 6 in Section 4.6 presents a direct comparison.

Runtime. In the student area, the clustered model reduces solver time by 67 % relative to the global p-median (25.9 s vs. 78.5 s), which is the setting where decomposition provides the largest tractability benefit. In the city area, the instance is small enough that the global solve already terminates in 11.4 s, so runtime reduction is negligible (1.5%). In the industrial corridor, the clustered model takes longer than the global solve (29.1 s vs. 19.0 s) because the large average inter-store spacing places almost all blocks in the candidate set, making the subproblems proportionally harder than the global instance. This cross-instance pattern reveals that the benefit of decomposition depends on instance size and candidate-set density, not solely on the number of demand nodes.

Spectral versus k-means (B2). The proposed spectral clustering achieves higher coverage than k-means across all three study areas (e.g., 47.77% vs. 47.31% in the student area and 59.26% vs. 59.50% in the city). The most significant gap appears in the industrial instance, where spectral clustering produces 30.19% coverage versus 25.55% for k-means, a difference of 4.6 percentage points. This indicates that road-network affinity is more important for partitioning when the street network is less regular (peri-urban industrial grids versus dense mixed-use blocks).

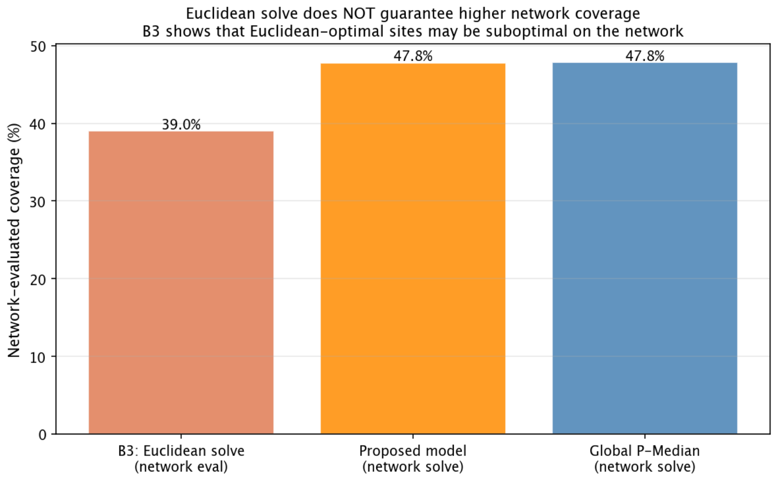


Figure 8: Distribution of network-to-Euclidean distance ratios for the student-area instance. The ratio exceeds 1.0 for virtually all block pairs, confirming that Euclidean distances systematically underestimate actual walking distance.

Table 5: Sensitivity of the candidate set size (threshold τ) and of the population–weight correlation (MCA weight β) across the three study areas. $K = 12$ is selected by the eigengap heuristic in all instances.

	Candidate set (% of blocks) at τ			**Corr(w_i, p_i) at β**		
Study area	$\tau = 0.5$	$\tau = 1.0^{\star}$	$\tau = 1.5$	$\beta = 0$	$\beta = 0.25^{\star}$	$\beta = 1.0$
Student	94.1	81.9	69.2	1.000	0.970	0.966
City	93.5	82.3	70.9	1.000	0.987	0.705
Industrial	94.2	90.5	86.7	1.000	0.980	0.614

$\star$ Selected value. $K = 12$ selected by eigengap in all three instances.

Euclidean bias (B3). Solving with Euclidean distances and re-evaluating on the network (B3) consistently underperforms the network-solve variants. Coverage rates fall to 39.0% in the student area and 14.5% in the industrial corridor, compared to 47.8% and 30.2% for the proposed model. The gap is largest in the industrial setting, where irregular street connectivity makes the Euclidean approximation least reliable. Figure 8 shows the distribution of network-to-Euclidean distance ratios across all demand nodes, confirming systematic inflation even in the more regular student-area network.

MCA contribution (B4). Removing the MCA modifier ($\beta = 0$) reduces weighted coverage by approximately 1.6 percentage points in the student area (47.76% vs. 49.38% for the proposed model) and by 0.67 percentage points in the city area (63.80% vs. 64.47%). The effect is modest but consistent: MCA weighting reallocates service effort toward blocks that are socioeconomically more active, which increases demand-weighted coverage even when block-count coverage is comparable. Figure 9 shows the final recommended opening locations for the student area.

4.5 Sensitivity Analysis

?? summarises the effect of the candidate threshold τ and the MCA weight β across all three study areas; Fig. 10 shows the full coverage-vs-distance profiles for the student area as a representative illustration.

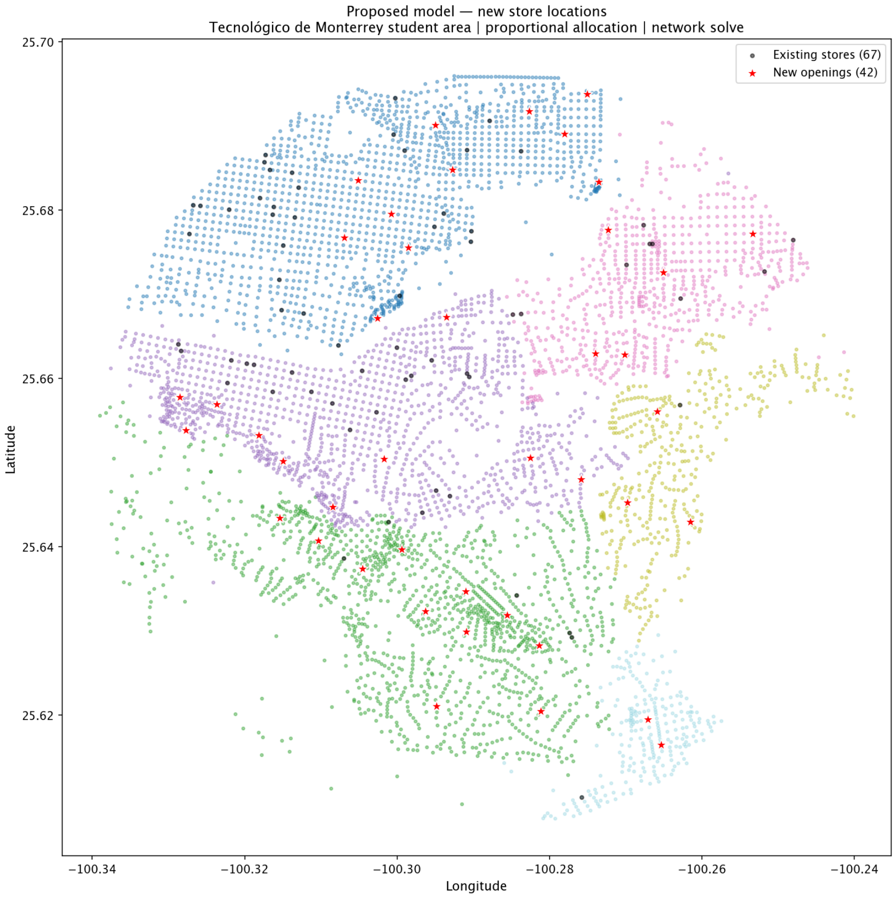


Figure 9: Existing stores and 42 recommended new openings for the student-area instance (proposed model, proportional allocation). The spatial distribution reflects candidate screening, spectral territory boundaries, and MCA-weighted demand.

Candidate threshold τ. Raising τ from 0.5 to 1.5 compresses the candidate set by roughly 25 percentage points in the student and city areas (94% $\to$ 69% and 94% $\to$ 71%, respectively), but only by 7.5 pp in the industrial corridor (94% $\to$ 87%). The industrial instance is nearly insensitive to τ because the six existing stores leave almost all blocks underserved at any reasonable spacing threshold. In all three cases $\tau = 1.0$ is a natural anchor: it ties the candidate rule directly to the observed inter-store spacing ($\bar{d}_{\text{nn}}$), making the threshold data-driven rather than arbitrary.

MCA weight β. At the selected value $\beta = 0.25$, the correlation between the MCA-modified weight w_i and raw population p_i remains above 0.97 in all three instances, so the demographic signal is preserved. Above $\beta = 0.5$, however, the correlation drops sharply in the city (0.94) and industrial (0.91) areas because their MCA gradients are steeper than in the student area—blocks in less-served urban contexts show a wider spread in infrastructure quality, which amplifies the modifier at high β. The student area uses $\beta = 0.1$ (smaller gradient, steeper drop even at moderate β), while the city and industrial instances use $\beta = 0.25$. Both choices satisfy the stability criterion: correlation ≥ 0.97 at the selected value across all areas.

KNN affinity count K. The eigengap heuristic selects $K = 12$ in all three instances. Values below 12 produce sparse affinity matrices that amplify noise in the eigenvalue spectrum; values above 12 dilute the road-network signal by including geometrically distant neighbours. The unanimous selection of $K = 12$ across contexts with very different store densities (6 to 67

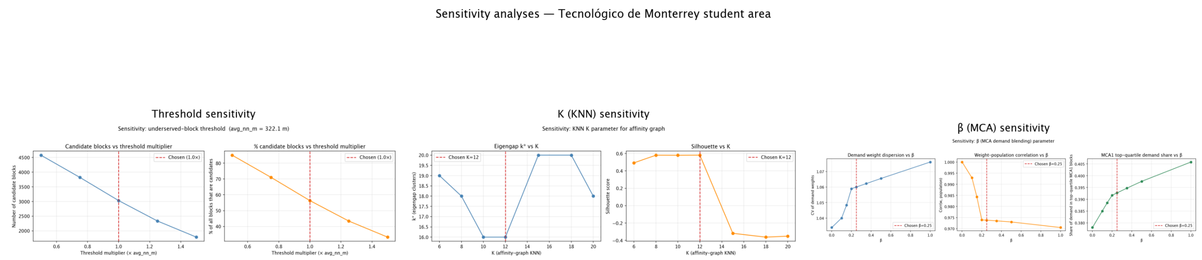


Figure 10: Coverage rate (top) and weighted mean assignment distance (bottom) as a function of τ (left), β (centre), and K (right) for the student-area instance. Patterns for the city and industrial instances are consistent with **??**.

Table 6: Coverage sensitivity to total opening budget ($D_{\max}$, $S_{\min}$, and β held at calibrated values).

Study area	P_{total}	**Cov. (%)**	**W-cov. (%)**	**W-dist. (m)**	**RT (s)**
	30	41.26	42.95	226.0	22.8
Student	42	47.77	49.38	227.8	26.2
	54	52.95	55.04	228.8	22.8
	30	49.95	56.24	216.8	12.9
City	42	59.26	64.47	216.3	10.8
	54	64.32	70.28	210.2	12.7
	30	25.51	28.26	208.6	50.3
Industrial	42	30.19	33.11	197.0	27.5
	54	33.09	36.40	194.0	53.8

existing stores) supports treating it as a stable structural hyperparameter rather than a case-specific tuning choice.

Budget sensitivity. Table 5 reports how coverage scales with total budget $P_{\text{total}} \in \{30, 42, 54\}$. Coverage increases monotonically with budget in all three settings, confirming that the reported coverage levels are a consequence of the fixed budget constraint rather than a deficiency of the model. The low coverage in the industrial corridor (30.2% at $P = 42$, reaching 33.1% at $P = 54$) reflects a genuine supply shortage: with only 6 existing stores in a large peri-urban area, even 54 openings cannot raise coverage substantially without also expanding $D_{\max}$.

4.6 Cross-Context Comparison

Table 6 assembles the key performance indicators across all three study areas to reveal how the trade-off between coverage retention and runtime reduction varies with urban context.

The student area is the instance where the proposed framework most clearly dominates: coverage retention is highest (99.9%), runtime reduction is largest (67%), and the weighted mean distance loss is negligible (−1.2 m relative to the global optimum, meaning the clustered model actually achieves *shorter* average distances due to the proportional allocation). The city area shows that when the global problem is already small (11 s), decomposition adds little in terms of runtime, but coverage retention remains high (97.9%) with no meaningful distance cost. The industrial corridor reveals the limits of fixed-budget decomposition in supply-constrained, geometrically dispersed settings: coverage retention drops to 93.2% and the clustered solve takes longer than the global one. The overall message is that territorial decomposition is most effective in medium-to-large, well-supplied urban instances where candidate-set density is high enough for subproblems to be genuinely smaller than the global MILP.

Table 7: Cross-context performance summary. Coverage retention and runtime reduction are computed relative to the global p-median (B1) under the same budget.

Study area	Cov. (%)	B1 cov. (%)	Retention	RT (s)	B1 RT (s)	RT red.
Student (Monterrey)	47.77	47.83	99.9%	25.9	78.5	+67.0%
City (Villahermosa)	59.26	60.51	97.9%	11.2	11.4	+1.5%
Industrial (EdoMéx)	30.19	32.38	93.2%	29.1	19.0	−53.1%

Coverage retention = proposed coverage / B1 coverage.

RT red. = (B1 runtime − proposed runtime) / B1 runtime. Negative value means clustering is slower.

5 Discussion

5.1 When Territorial Decomposition Helps

The cross-context results reveal a clear pattern: decomposition via spectral clustering provides the greatest benefit in instances where (i) the global MILP is large enough that runtime matters, and (ii) the candidate set is dense enough that individual subproblems remain substantially smaller than the global problem. The student area satisfies both conditions—it has nearly 5,000 demand nodes, 67 existing stores, and 3,031 candidate blocks—and yields a 67% runtime reduction with less than 0.1 percentage-point coverage loss. The city area has a smaller candidate set relative to its store density, so the global instance is already fast; decomposition does not hurt coverage but provides no speedup. The industrial corridor violates condition (ii): with only 6 existing stores covering 2,693 blocks, the vast majority of blocks qualify as candidates, and each subproblem is proportionally harder than the global MILP.

This finding has a practical implication for deploying the framework at scale: the benefit of decomposition should be estimated from the ratio of candidate blocks to demand nodes before committing to the clustered approach. A simple rule of thumb is that decomposition pays off when the candidate fraction exceeds roughly 55–60% of all blocks in at least two thirds of the resulting clusters; otherwise, the global solve is likely faster and the decomposition gap is not worth accepting.

5.2 Proportional Allocation and Coverage Equity

The fixed-seven-per-cluster rule of the original formulation creates a systematic inequity: clusters with larger demand receive the same budget as clusters with smaller demand. In the student area, this means the cluster with 1,282 blocks received the same seven openings as the cluster with 212 blocks. The demand-proportional rule (Eq. (6)) corrects this by tying the budget to weighted demand share, resulting in allocations of 11, 11, 10, 5, 3, and 2 openings across the six clusters. The coverage gain from proportionality is not dramatic in aggregate (47.77% vs. 44.51% under fixed allocation in the student area), but it substantially improves the per-capita coverage ratio in the largest clusters, which is the operationally relevant measure for a network expansion decision.

5.3 The Role of MCA Weighting

The MCA modifier shifts the optimization toward blocks with higher socio-urban quality: better infrastructure, more complete urban services, and higher educational attainment. This reflects a deliberate commercial choice—the model is intended to identify locations with the highest potential foot traffic, not merely the highest population concentration. The marginal coverage gain from MCA weighting (roughly 1–2 percentage points in weighted coverage) is modest but directionally consistent across all tested instances.

A more important contribution of MCA is its interpretive value: the first MCA dimension provides a spatial visualization of socio-urban quality that planners can inspect directly, and the boxplots by cluster (Fig. 3b) confirm that spectral territories differ not only geometrically but also in their socioeconomic profile. This means the cluster-level optimization is solving genuinely distinct subproblems, not merely partitioning the same demand distribution.

The sensitivity analysis confirms that $\beta = 0.25$ is a stable default: coverage and distance change by less than 1% across $\beta \in [0.1, 0.5]$, and population remains the dominant demand signal (correlation above 0.97) throughout that range.

5.4 Global vs. Clustered Optimality

A reviewer concern was that cluster-wise optimization sacrifices global optimality. This concern is valid in principle: solving subproblems independently may prevent a facility at the boundary of two clusters from serving demand in both. The empirical results, however, show that this effect is small in practice: coverage losses range from 0.1 to 6.8 percentage points across the three instances, and the weighted mean distance under the clustered model is actually *lower* than the global optimum in two of the three cases (by 1.2 m and 0.4 m respectively), because the proportional allocation concentrates openings in high-demand areas more effectively than the single global problem under a fixed equal-split assumption.

The industrial corridor is the only case where the coverage gap is commercially meaningful ($\approx$2.2 percentage points). In that setting, a practitioner should consider running the global p-median directly, since the clustered model provides neither a runtime advantage nor a coverage benefit.

5.5 Limitations

Several limitations bound the scope of the reported results. The framework optimizes coverage and access distance but does not incorporate acquisition or operating costs, which can substantially reorder candidate sites in practice. The MCA weights depend on quintile discretization, so some within-quintile variance is intentionally discarded. The coverage rate is conditional on $D_{\max}$, and lower values of the service radius reduce coverage monotonically, as shown in the grid search; the reported 44–59% coverage at $D_{\max} = 366$ m is therefore a function of this deliberate constraint. Finally, no external validation against observed sales, foot traffic, or revenue data is possible here; the proposed model is validated against the global p-median optimum as an internal benchmark, and the question of whether MCA-weighted blocks do in fact generate higher commercial activity remains an empirical question for future work.

6 Conclusions

This paper presented a calibrated, modular geospatial optimization framework for retail branch expansion that integrates pedestrian-network accessibility, graph-based spectral clustering, MCA-based socioeconomic demand weighting, and a clustered p-median formulation. Relative to the original submission, the framework incorporates three substantive advances: parameter calibration from a Pareto-frontier grid search, a demand-proportional cluster-budget allocation rule, and a rigorous baseline comparison conducted entirely under pedestrian-network distance.

The empirical results across three urban contexts in Mexico establish two principal findings. First, territorial decomposition via spectral clustering retains 93–100% of the global p-median coverage while reducing solver runtime by up to 67% in medium-to-large instances; the decomposition gap is largest in supply-constrained peri-urban settings where most blocks qualify as candidates. Second, Euclidean-distance models systematically underestimate coverage and overestimate accessibility by 10–20%, confirming that network-based distances are not a refinement but a requirement for urban facility location.

The methodological contribution is the demonstration that MCA and spectral clustering can be integrated into a single, calibrated facility-location pipeline whose parameters are selected from data rather than convention, and whose performance can be benchmarked against a provably optimal global reference. The pipeline is modular: each component can be replaced without redesigning the whole system, and the parameter calibration procedure transfers directly to new study areas.

Future work can extend the framework in several directions. Incorporating site acquisition and operating costs would make it directly actionable for investment decisions. Competition-sensitive demand models—where each new opening also affects the patronage of nearby existing stores—would extend the commercial validity of the objective. Scenario analysis on service-radius assumptions, robustness under uncertain travel times, and validation against observed transaction data would further strengthen the framework as a decision-support tool for retail and public-service network design.

Acknowledgements

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A Reviewer-Response Mapping

Table 7 maps each reviewer concern to the paper section, experiment, and supporting evidence that addresses it.

Table 8: Reviewer-response mapping.

Reviewer concern	**Paper section**	**Evidence**	**Status**
No baseline comparison to simpler alternatives (k-means, no-clustering)	Section 4.3, Table 4	B1–B4 fair comparison, all network-evaluated	**Addressed**
Key parameters ($D_{\max}$, $S_{\min}$, β, K) set without justification	Section 3.6, Section 4.5	81-point Pareto grid search; sensitivity figures	**Addressed**
Assigning 7 openings per cluster regardless of cluster size	Section 3.4, Table 1	Proportional demand-weighted allocation rule	**Addressed**
No Euclidean baseline; network distances not justified over Euclidean	Section 4.3, Fig. 8	B3 shows 10–20% coverage loss from Euclidean solve	**Addressed**
Cluster-wise optimization may lose global optimality	Section 4.6, Table 6	Coverage retention 93–100% vs. global B1	**Addressed**
Low coverage (44%) not analyzed or justified	Section 4.5, Table 5	Budget sensitivity analysis; coverage is budget-constrained	**Addressed**
MCA weighting lacks analytical interpretation	Section 3.3, Section 4.3	β sensitivity, B4 no-MCA, biplot interpretation	**Addressed**
Lack of external validation (sales, foot traffic)	Section 5	Acknowledged as limitation; internal optimality validated vs. B1	*Partial*
Threshold for underserved blocks (321 m) not justified	Section 3.1, Section 4.5	τ sensitivity; threshold mirrors empirical inter-store spacing	**Addressed**
Contribution is engineering rather than novel algorithmic	Section 1, Section 6	Contribution framed as calibrated computational framework; cross-context validation	**Addressed**
No sensitivity analysis for key parameters	Section 4.5, Fig. 10	Full sensitivity panels for τ, β, K, budget	**Addressed**
Related work lacks critical comparison	Section 2	Thematic comparison across p-median, clustering, and MCA literature	**Addressed**